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Topic : Content Beyond Syllabus

Q.1 Explain following variations of gradient descent algorithm and compare them with Adam.

a) AMSGrad

⇒ - Adaptive Moment Estimation with ~~Logn~~ Long-term Memory is a variant of ADAM that fixes an issues where Adam may have poor generalization due to its adaptive learning rate scheme

- Similar to Adam, but it maintains a maximum of past squared gradients instead of an exponentially decaying average.
- This prevents rapidly decreasing learning rates, which can cause convergence issues.

b) AdaBelief

⇒ - Adaptive Belief Optimization is an improved version of Adam that adapts step size based on how much the gradient changes over time.

- Similar to Adam but measures the belief in the gradient by tracking its variance.
- If the gradient changes a lot, learning rate is reduced.
- If gradient remains stable, a larger learning rate is maintained.

c) AdaDelta

⇒ - Adaptive Delta is designed to overcome the limitations of AdaGrad, which suffers from a vanishing learning rate problem.

- Instead of accumulating all past squared gradients, it keeps an exponential moving average.

- This prevents the learning rate from shrinking to near zero over time.

d) AdaBound

⇒ - Adaptive Bound is an improvement over Adam that gradually transition from an adaptive method to SGD.

- Starts like Adam but gradually bounds the learning rate between a min and max threshold.

- This prevents extreme fluctuations in learning rates.

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- Comparison :

Optimizer	Key Feature	Problem Solved	Best For
Adam	Adaptive learning rate using momentum.	Fast convergence but poor generalization.	General-purpose deep learning.
AMSGrad	keeps the max of past squared gradients	Prevents Adams convergence issues.	More stable learning rates
AdaBelief	Measures belief in gradient	Better stability and generalization	Faster and stable training
AdaDelta	Exponential moving average of squared gradients.	Prevents vanishing learning rate.	No need for manual tunning
AdaBound	Learning rate bounded between min and max values.	Avoid extreme fluctuations.	Combining Adam + SGD benefits

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Q.2 Explore ChatGPT and Deepseek models to understand what deep learning techniques are applied in these models. Write a technical article encompassing details of the working of the models.

⇒

Introduction

Large Language Models (LLMs) like ChatGPT and Deepseek are advanced AI systems that generate human-like text based on deep learning techniques. These models utilize architectures built on Transformers networks, leveraging self-attention mechanism and vast amounts of data to perform natural language understanding and generation tasks effectively.

1. Core Deep Learning Techniques

Both ChatGPT and Deepseek rely on deep learning techniques that enhances their ability to understand, generate and process natural language. The key techniques include:

1.1 Transformer architecture.

Transformer architecture is the foundation of LLMs. It replaces traditional recurrent neural networks (RNNs) and Convolutional neural networks (CNNs) with the self-attention mechanism, allowing parallel processing of input data. The model consists of encoder-decoder or decoder-only models.

1.2 Pretraining and Fine-tunning

Both models are pretrained on massive datasets using self-supervised learning with tasks like masked language modeling and casual language modelling. ChatGPT is fine-tuned using Reinforcement Learning From Human Feedback, optimizing responses based on human preferences, whereas Deepseek finetunes for domain-specific tasks

1.3 Tokenization.

Text input is tokenized into subword units using Byte Pair Encoding. This helps handle rare words efficiently and reduces the vocabulary size needed.

1.4 Optimization techniques.

Uses AdamW optimizer for stable convergence. Gradient checkpointing reduces memory usage in training large-scale models.

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2. ChatGPT vs DeepSeek

While both models are based on Transformers, they have distinct focuses :

ChatGPT	DeepSeek
1) Decoder-only Transformer.	1) Decoder-only transformer
2) General internet corpus, books and conversation.	2) Specialized technical & research-focused datasets
3) For e.g., Conversational AI, Code assistance,	3) For e.g., Technical and research document generation

3. Conclusion

ChatGPT and DeepSeek leverages cutting-edge deep learning techniques, including transformer models, pretraining, fine-tunning. While ChatGPT is optimized for broad, conversational AI tasks, DeepSeek is more specialized in domain-specific applications

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